Detecting Covert Optical Signaling in Camera Footage: A DSP Pipeline for Blink-Count and Brightness Channels

Samet Yılmaz Temel
0rce-Labs
https://github.com/0rce-Labs/lumenradar

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Abstract

Consumer software can modulate a screen or status LED so that a nearby camera receives data a human supervisor misses: answer codes encoded as blink counts, brightness pulses, or sub-perceptual luminance oscillation. The attack side of this channel family is well studied; the *detection* side is not — prior work explicitly leaves camera-based monitoring unimplemented. We present LumenRadar, an open-source detector that analyzes ordinary video with a three-detector DSP pipeline (narrowband periodicity, burst pulse-count structure, session-baseline comparison) fused with camera-artifact rejection into an explainable verdict. On synthetic calibration through an mp4 codec round-trip, the pipeline detects 2 Hz modulation down to ∼4 gray levels peak-to-peak at 30 fps with zero false positives, and recovers blink-count symbol sequences exactly. On a real-hardware screen-recording dataset (Retina 2880×1800, variable frame rate), all three positives are detected with no false positives after three genuine real-world defects — screen-rendered beacons beyond LED-size heuristics, low-duty lead-in timing, and VFR spectral leakage — surfaced by the data and were fixed

1 Introduction

Screen- and LED-control software in the exam-proctoring evasion space now ships covert answer channels: menu-bar icons, Caps-Lock LED blink counts, keyboard backlight pulses, and screen-brightness flashes, marketed as undetectable in video-conferencing proctoring. The optical covert-channel literature covers the attack side extensively [1, 2, 3, 4, 5], but detection remains an open gap: [2] explicitly describes camera-based monitoring as "nontrivial" and leaves it unimplemented, while receiver-side work solves *decoding* — the attacker's problem — not *detection*.

This paper contributes:

1. a three-detector DSP pipeline fused with artifact rejection into an explainable verdict (§4);
2. the first open-source detector targeting blink-count symbol modulation in the proctoring context;
3. a perceptibility-based calibration quantifying the detection floor (∼4 gray levels p-p, 30 fps, mp4 round-trip) and an open dataset protocol with integrity-hashed manifests (§5).

Scope: detection only. The accompanying repository contains no transmitter and no answer-generation path; its calibration beacon exists solely to label positive-class samples on the defender's own hardware.

2 Related Work

3 Threat Model

Attacker: an app or malware controlling an LED or screen region on the supervised device, plus a receiver camera with line of sight. **Defender:** holds only a camera stream (e.g. the proctor webcam); no access to the supervised device. **Signal classes:** (1) blink-count symbols (1–4 pulses = choice A–D, repeated), (2) periodic brightness pulses, (3) sub-perceptual narrowband oscillation. PWM-rate channels beyond camera Nyquist are out of scope.

Work	Channel	Direction	Gap we fill
LED-it-GO [1]	HDD LED OOK, 4 kbit/s	attack	tape/jam only
BRIGHTNESS [2]	screen brightness	attack	detection "nontrivial"
VisiSploit [3]	sub-perceptual images	attack	—
HiLight [4]	translucency FSK	attack (decode)	DSP adapted defensively
Sensors'23 [5]	brightness 2-PAM	attack (decode)	decoding $\neq$ detecting

Table 1: Positioning against the optical covert-channel literature.

4 Method

Per clip: ingest $\rightarrow$ LED ROI localization (small saturated stable blob; manual override; full-frame fallback) $\rightarrow$ ROI luminance series plus a control series (frame mean with the ROI/ring excluded) $\rightarrow$ three detectors $\rightarrow$ artifact classification $\rightarrow$ fused verdict with full evidence.

M1 — band periodicity (0.5–15 Hz). Hann PSD with three gates: absolute amplitude $A = 4\sqrt{P}/N \geq 0.5$ gray (perceptibility; also kills out-of-band leakage skirts), peak-to-local-floor ratio ≥ 8, and peak above the Gumbel noise-max expectation $2(\ln n + \gamma)$. Harmonic confirmation against the global floor.

M2 — burst pulse-count. Threshold segmentation $\rightarrow$ intervals $\rightarrow$ bimodal split at the largest jump (a median split collapses when single-count symbols dominate) $\rightarrow$ symbol counts; detected iff counts repeat AND intra-gap CV ≤ 0.6 AND inter-gap CV ≤ 0.35 AND gap contrast ≥ 2. Aperiodic activity (e.g. HDD LEDs) fails the pacing test.

M3 — session baseline. The first 2 s form a beacon-free profile; candidate windows are compared by smoothed PSD ratio with $z \geq 5$ (log-ratio $\sigma = \sqrt{2\psi'(5)}$) AND amplitude ≥ 0.5 gray — near-zero-energy baselines make raw ratios explode on codec/quantization noise.

Artifacts. Auto-exposure hunting: ROI/control lockstep ($r \geq 0.85$) *with* comparable magnitude (control/ROI std ≥ 0.25); correlation alone is fooled by mp4 block-coupling (measured $r = 0.86$–0.98 with ≈ 0 control amplitude). Mains $2f$ aliases are flagged only when Nyquist covers 100/120 Hz. Full-frame mode derives a spatial control from the border ring.

5 Dataset Protocol

Clips follow a `pos_/neg_` naming contract; a SHA-256 manifest pins integrity; the evaluator runs the identical analysis path as the CLI. Real-hardware results (6 clips: 3 positives, 3 negatives, Retina 2880×1800 screen recording, 24–52 fps VFR): **3/3 positives detected** (2×signal_detected as blink_count with symbol intervals recovered, 1×suspicious periodic at the true 1.8 Hz), **0 false positives**. Capturing these numbers surfaced three real-world defects that synthetic data alone could not: the calibration beacon renders as a ∼300 px solid bright square beyond the small-LED ROI heuristic (fixed with a class-aware beacon-square scan), low-duty blink patterns need a wider ROI probe window (first ON frame at 0.65 s), and variable-frame-rate capture turns a strong tone's spectral skirt wide enough to defeat local-floor ratios (fixed with a strong-amplitude+harmonic bypass, and a sustained-power gate that rejects step-edge transients — measured 0.05 vs 1.00 sustained-fraction separation). Real-hardware recordings are not redistributed; a SHA-256 manifest and all generation/analysis scripts are provided for independent verification, and clips are available from the author on reasonable request (see `docs/DATASET-POLICY.md`). Remaining for the journal version: scaled-up dataset (2 devices × 3 lighting, ≥60 clips) and formal ROC.

6 Synthetic Calibration (preliminary)

Config: 30 fps, 6 s clips, structured background scene, 16×16 LED, mp4v round-trip, 5 seeds per amplitude, 10 clean seeds. Detection floor (2 Hz sine, TPR 5/5 from): **4 gray levels p-p**; clean FPR **0/10**. Frequency sweep (∼76 gray levels p-p, 3 seeds/frequency): TPR 3/3 at every frequency from 0.5 to 14 Hz with the peak recovered at the true frequency; at 15 Hz the mp4 codec itself destroys the signal (amplitude drops to ≈0.1 gray), so the practical band edge is codec-bound, not detector-bound. End-to-end: clean/noisy $\rightarrow$ clean; blink pattern $(1, 1, 2, 1, 2, \dots)$ $\rightarrow$ signal_detected with symbol counts recovered exactly after codec; whole-frame oscillation $\rightarrow$ camera_artifact. Runtime: ∼12–16× realtime on an ARM aarch64 core (30 s clip in 2.2 s). All numbers are synthetic-only; real-camera data are expected to shift the floor.

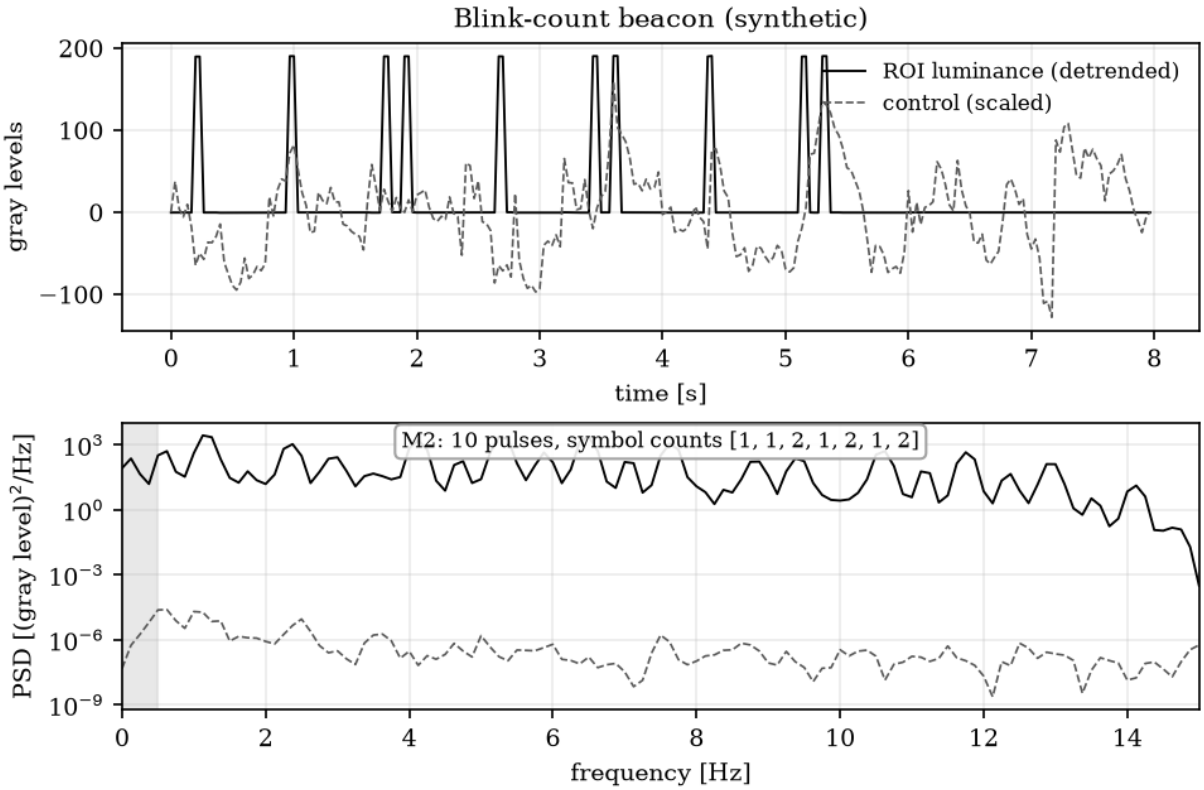


Figure 1: Blink-count beacon (synthetic, codec round-trip). Top: detrended ROI luminance (black; 10 pulses in symbol groups) vs. the ROI-excluded control (dashed gray, scaled). Bottom: Welch PSD of both channels; the shaded strip is the excluded < 0.5 Hz band. The annotation reports the M2 detector output: 10 pulses, symbol counts $[1, 1, 2, 1, 2, 1, 2]$ — recovered exactly through the codec.

7 Countermeasures

For flagged sessions: physical LED-covering policy, brightness-locked displays, proctor-cam exposure lock, and continuous session-baseline monitoring. Detector output is evidence (JSON), never an automated accusation.

8 Threats to Validity

Synthetic scenes lack realistic illumination dynamics; the beacon is machine-timed while human-driven codes jitter (M2 CV thresholds may need loosening); adaptive adversaries can randomize duty cycles to defeat pacing; lossy webcams compress away sub-0.5-gray signals — which also raises the attacker's required modulation depth, favoring the defender.

9 Conclusion and Roadmap

LumenRadar closes a documented gap: an open, explainable detector for covert optical signaling in ordinary camera footage, with a calibrated synthetic floor and a reproducible dataset pipeline. Next: real-hardware ROC, arXiv preprint, conference submission.

References

[1] Mordechai Guri, Boris Zadov, Eran Atias, and Yuval Elovici. LED-it-GO: Leaking (a lot of) data from air-gapped computers via the (small) hard drive LED. *arXiv preprint arXiv:1702.06715*, 2017.

[2] Mordechai Guri, Matan Monitz, and Yuval Elovici. BRIGHTNESS: Leaking sensitive data from air-gapped workstations via screen brightness. *arXiv preprint arXiv:2002.01078*, 2020.

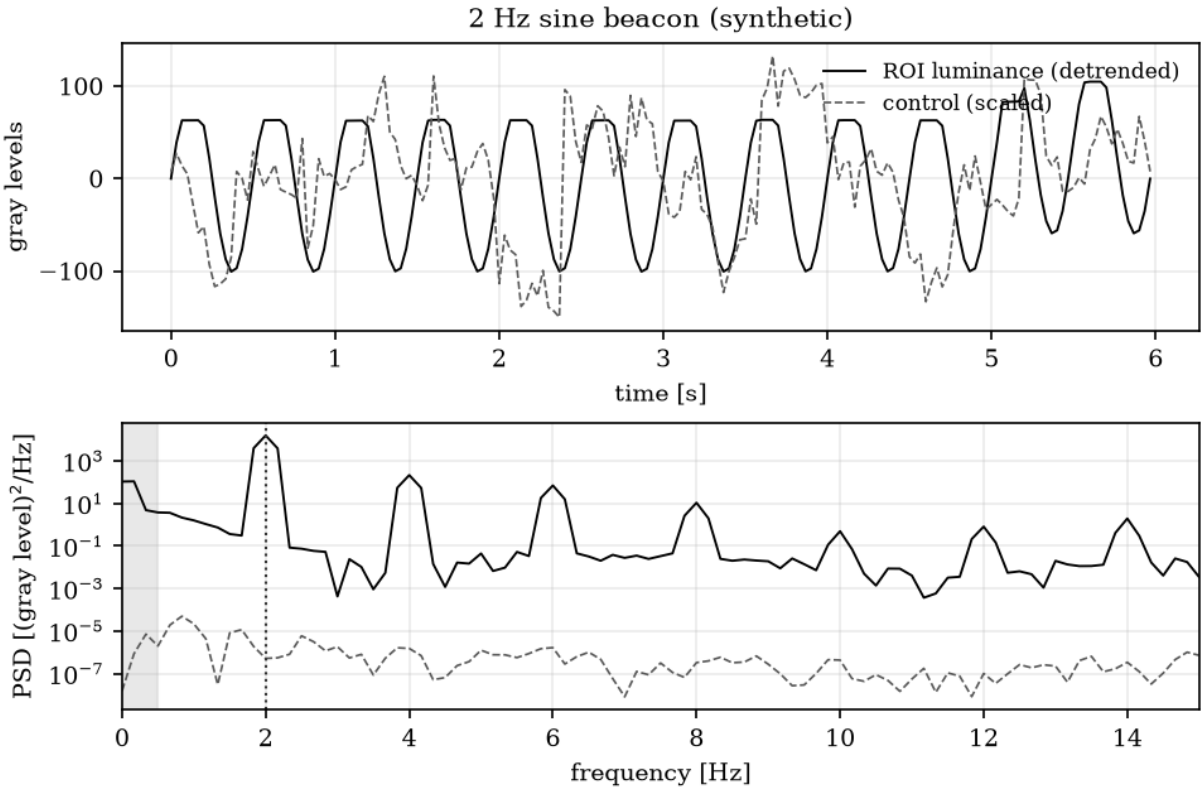


Figure 2: Narrowband beacon (synthetic): 2 Hz modulation is ∼8 orders of magnitude above the control channel in the Welch PSD; the dotted marker is the M1 peak.

[3] Mordechai Guri, Dima Bykov, Boris Zadov, and Yuval Elovici. VisiSploit: An optical covert-channel to leak data through an air-gap. *arXiv preprint arXiv:1607.03946*, 2016.

[4] Tianxing Wang, Yang Liu, and Athanasios V. Vasilakos. HiLight: Hiding bits in pixel translucency changes. In *Proceedings of the 22nd Annual International Conference on Mobile Computing and Networking (MobiCom)*, 2016.

[5] Przemyslaw Mazurek and Dawid Bak. Video sequence segmentation based on K-Means in air-gap data transmission for a cluttered environment. *Sensors*, 23(2):665, 2023.